Luke Zhuang

San Francisco | lwzhuang@calpoly.edu | 415.837.8686 | https://luke7787.github.io/Portfolio/

EDUCATION

California Polytechnic State University, San Luis Obispo **Expected March 2025**
Bachelor of Science Degree in Computer Science

RELEVANT COURSEWORK

Data Structures, Object-Oriented Programming, Software Engineering, Design and Analysis of Algorithms, Computer Security, Artificial Intelligence, Systems Programming, Operating Systems, Databases, Computer Organization

TECHNICAL SKILLS

Cloud & DevOps: AWS, Azure, GitHub Actions, Continuous Deployment, Cloud Computing
Frameworks & Libraries: React.js, Node.js, Express.js, Sass, JWT, Axios, Bcrypt, Figma
Programming Languages: JavaScript, TypeScript, Python, Java, C, HTML5, CSS3, C++
Databases: MongoDB, SQL

WORK EXPERIENCE

Software Engineer, California Polytechnic State University **January 2025 – Present**

- Leverage AI and data analytics to predict missing person locations by analyzing key factors such as elapsed time, travel distance, weather, terrain, and age.
- Enhance search efficiency by mapping high-probability hot spots, improving response time and rescue success rates.

Front Desk Staff, Valencia Apartments, CA **January 2025 – Present**

- Assisted over 100 tenants with inquiries, provided solutions, and coordinated with management to address tenant concerns.

Barista & Donut Shop Associate, SloDoCo, CA **May 2024 – September 2024**

- Delivered exceptional service by brewing high-quality coffee, serving customers with a friendly attitude, and maintaining a clean, organized work environment, contributing to a 99% customer satisfaction rate.

TECHNICAL PROJECTS

Inventory Management App **July 2024 – September 2024**

- Implemented secure user authentication with encrypted passwords, reducing security vulnerabilities by 32%
- Deployed application assets and data on Amazon S3, improving data retrieval speed by an estimated 26%.
- Integrated TypeScript to enforce type safety, enhancing code reliability and reducing bugs by 21%.

Blackjack **January 2024 – June 2024**

- Developed a fully responsive JavaScript Blackjack game compatible with desktop and mobile devices.
- Authentic gameplay features, including hand splitting, bet doubling, and special Blackjack payouts, increasing user engagement by 40%.

Server and Client **September 2023 – December 2023**

- Developed a lightweight web server in C supporting a targeted subset of HTTP methods for efficient request-response handling, improving latency by 15%.
- Implemented child process forking to handle up to 50 concurrent client requests, improving scalability.

Matrix Multiplication: **September 2023 – December 2023**

- Parallelized matrix multiplication using SIMD instructions and multithreading, achieving a 48% performance improvement.